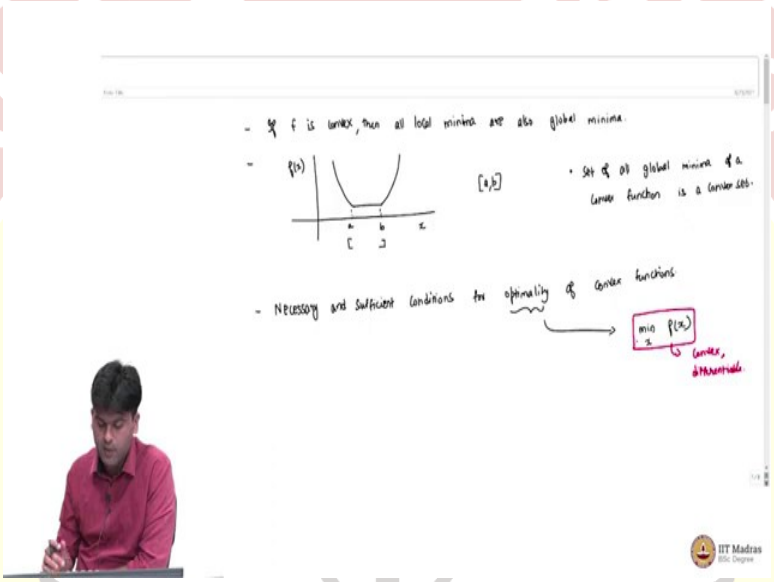


IIT Madras

ONLINE DEGREE

Machine Learning Foundations
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Properties of Convex Functions

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Hello, everyone. Welcome back. Today we will continue our discussion about convex functions and some of their properties, and then we will try to see how we can use convexity as an important property to set up machine learning problems and we will see an example for that as well. So, this would be the goal going forward.

Just to remind ourselves, so we have so far seen that if f is a convex function, let me remind you, if f is convex, when you say if f is convex it means that I mean f is a convex function, then all local minimum are also global minimum. So, this is something that we proved last time. One of the points that I just want to highlight is that this does not mean anything about that there being only one global minima. It just means that every local minima is also a global minima, but then the global minima may not be unique.

So, take a simple example of a function which looks like this, x, f of x, now this is a convex function. And if you look at the global minima of this function, it is every point in the interval a comma b. So, now this means that there could be multiple global minima. So, it is also a fact that

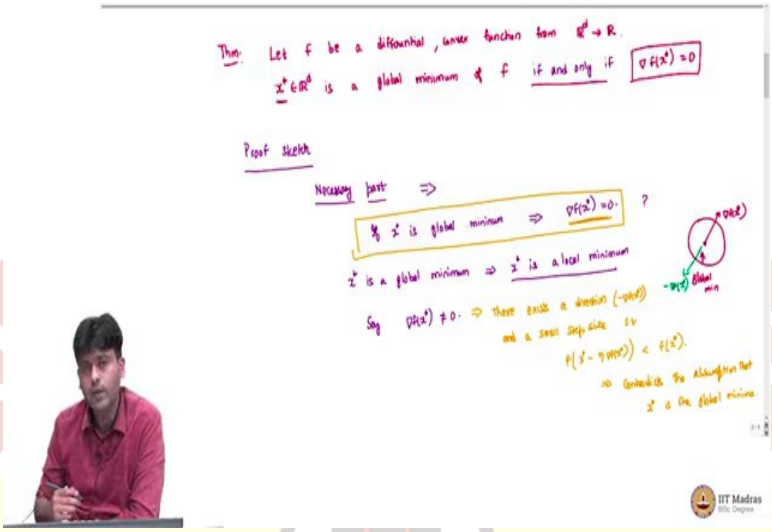
if you have multiple global minima then necessarily all of them should be in an interval if your function is from real to real or the set of all global minima themselves will form a convex set, set of all global minima of a convex function is a convex set.

So, try to prove this. This is not too hard. And in, for functions which are real to real, so this is simply saying that the set of all global minimum must necessarily be an interval. So, that is all this is saying. Now, we want to understand, go forward and understand, so let us say if a function is convex, I know that the global or local minimas are global minima, but then what is the condition that determines minima local or global in this case. So, that is what we are going to start by looking at today.

In other words, what are the necessary and sufficient conditions for a differentiable function which is also convex to have a global minimum? So, we are going to look at necessary and sufficient conditions for optimality of convex function. So, when I say optimality, I mean that our goal is to minimize over $x, f(x)$ and we are going to assume f itself is convex and we will also assume f is differentiable.

Now, if I give you a differentiable convex function, now what can I say about properties that the optimal solution satisfies? That is the question we are asking. So, in other words, if I give you a point and say that this point satisfies a certain condition, then can we conclude that this necessarily, this is necessary and sufficient for this point to be an optimal point, which means that in this case optimality means the minimizer of the function. So, how can we do that? Let us put down a theorem for this.

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Maybe it will go to the next page. So, here is the theorem. Let, f be a differentiable convex function from, we will simply assume, we will always think about convex functions as Rd to R. So, some function which is Rd to R it is differentiable and it is also convex. When I say differentiable, it means that at any point you can compute the gradient of that function. That is what differentiability means.

Now, then x* a point in Rd is the global, is a global minima of f and now this is a necessary and sufficient condition, so we will say if and only if del of f(x*) is 0. What are we saying? we are saying that, hey, I give you a convex function and it is differentiable, then if you say x* is a global minimum, then del of x*, f(x*) has to be 0. Also, if there is a point such that del of f(x*) is 0, then necessarily that point is the global minimum. So, this is an if and only if condition, which means that it completely, this condition can completely characterizes the optimality of convex functions.

We are going to give a proof sketch for why this is true. So, let us first do the, do one of these conditions, necessity conditions and then we will do the sufficiency conditions. So, the necessary part. So, if x* is a global minimum, then del of f(x*) has to be 0. So, this is the necessary part that is, if x* is the global minimum, we need to show, then this implies del of f(x*) = 0. Why is this true? So, we will do a proof sketch, not the exact proof, but I will give you a argument as to why this should be true.

Well, if x* is a global minimum, then, well, it necessarily also has to be a local minimum, so x* is also a local minimum. Which means what? So, I mean, I am just trying to use the local minimum because I want to use something that we already have seen. So, what does that mean to be x* is local minimum, which means that there is always, so every direction you can move, and then you will see that the function value is actually going up. That is what local minimum means.

So, there is a small ball around this point such that if you move in that any direction in within the ball, your function value actually increases. That is what it means by definition of a local minimum. Now, we know that if, let us say, now, del of f(x*) is not 0. So, you are at some point which is, which you are assuming is a global minimum, so this is you are assuming as global minimum. Now, which means that no matter which direction I move in, I can move small enough amount such that I will increase the function.

But if the gradient at this point is not equal to 0, then we know from our Taylor series analysis that you can always, let us say, the gradient was pointing in this direction, I can always move in the direction opposite to the gradient by a small amount and then decrease the function value. This we argued, when we were looking at Taylor series expansions and arguing why the negative of the gradient is a good direction to move in, in general, for gradient descent and so on. So, this is minus del of f(x*).

Now, so we will not formally rigorously argue this. We will just say this implies there exists a direction, in particular, you can move in the minus del of f(x*) direction and small step size such that x* minus eta del of f(x*) this function value so will be less than f(x*). So, you can move in that direction and strictly go become, the functions value will strictly become less than this point. But that contradicts. So, that this contradicts the fact that or rather the assumption that x* is the global optimum.

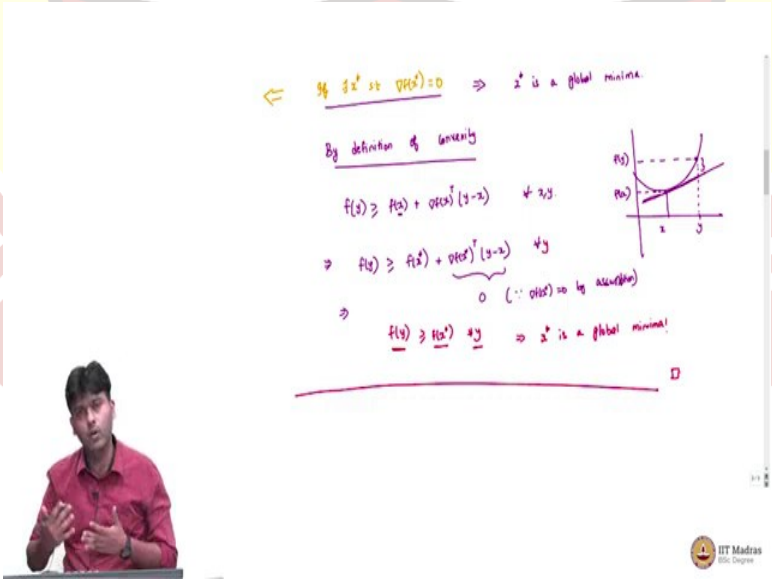
So, if it is a global optimum, then you should not be able to find a direction where you can move and strictly reduce the function. But if the gradient value is not 0 then you can move in that direction to reduce the function and so your assumption that the point was global minima to begin with was wrong. So, this means that if this proves the necessity part that if f(x*) is global minimum then necessarily del f(x*) is 0. Note that we have not used convexity at all. So, we did not argue that this is because the function is convex.

Now, where do we need convexity? We will need convexity to argue the other way. So, because it is not necessarily true that for any function if del of f(x*) is 0 then it means that x* is a global minimum. So, that is not necessarily true always, because the function could be concave function which is like the negative of a convex function where at the maxima you will still find del of f(x*) equal to 0 or there may be more complicated function where the function is, has a point where the gradient is 0, but then in one direction the function is increasing and the other direction the function is actually decreasing.

So, there are all sorts of things can happen when deal of f(x*) is 0 when you are not assuming anything about the structure of the function. So, this only becomes a necessity condition for a general function for x* to be the global minimum. If x* is global minimum then f of x, del of f(x*) is 0. So, this is a necessary condition.

In fact, this is a necessary condition for even a local minimum for a general function. Because local minimums are all global minimums, this becomes a necessary condition for global minimum for a convex function. But the other way in general is not true for general functions, but you will see why it is true for convex functions. So, what is the other way?

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The other way is the following. So, now we are going to prove the other way, which is the sufficiency, which means that if there exists x* such del of f(x*) equal to 0. So, if I tell you that there is a point where the gradient is 0, then this implies x* is a global minimum. Now, why would

this be true? Now, we, this is not, in general, true for any function. But we have to somehow argue that this is true when the function is convex. So how do we show this?

Now, here is where we use convexity. Now, let us use the definition of convexity, by definition of convexity, and remember there are several equivalent definitions of convexity and we are going to look at one specific definition which will be useful for us to prove this fact. One of the definitions was that, I take any two points x and y, and I, so let us call this x, let us call this y, and I look at the value of functions value at y, so this is f of y, this is f of x, I look at the functions value f of y, and this will always be greater, so there is always going to be greater than or equal to the linearization of the function at x. So, and we discussed this last time.

So, this is simply formal is saying that f of y is greater than or equal to f of x plus del of f of x transpose y minus x for all x comma y. I can take any two points and this holds. So, the tangent at x, so it is the tangent plane kind of lower bounds the function itself at any point. That is something very, very special about convex functions. This is an equivalent definition of convexity.

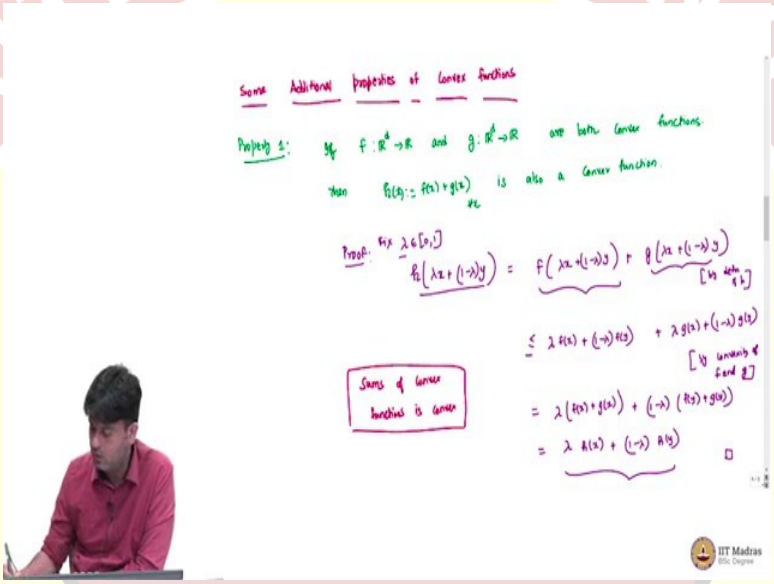
Now, what does this tell us, like, which, now this implies f of y, let us say, y is any y, but let us say x is x*, so this has to be greater than or equal to f(x*) plus the gradient of f(x*) transpose y minus x for all x. I am just applying this for the particular point x*, which I have assumed with satisfying the property that del of f(x*) is 0. So, now, this means that this simply implies because this guy is now 0 since del of f(x*) is 0 by assumption and that proves it.

So, now, what does imply? This finally implies f of y is greater than or equal to f(x*) for all, sorry, this has to be for all y, this is also for all y. That is it. So, now, what have we said? We have said that the functions value at any point y has to be greater than or equal to f(x*). Now, this implies x* is a global minimum. So, now this kind of proves the theorem that if x* has gradient 0, then its global minimum, and if x* is global minimum then the gradient is necessarily 0. So, gradient being 0 is a first order optimality condition for convex functions.

Why is this useful? This is useful because this is another way to say that, now, if you run a gradient descent algorithm on a convex function, well, the gradient descent algorithm will converge to a point where the gradient is of the function is actually 0. And now this is saying that such a point is necessarily a global optimum, which means that the gradient descent algorithm will actually find the global optimum for our functions.

So, last time, we argued saying that the gradient descent would find a local minimum and local minimums or global minimums and so gradient descent will find the global minimum. Now, we are making it more precise by saying that gradient descent will actually find a point for which the gradient is 0. It will converge to a point for which the gradient is 0 and that point is the global optimum. So, which is good, which means that all the algorithms that we have seen can actually be used to compute the global optimum of convex functions.

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Now, what we want to do next is see a practical example in machine learning, where you can set up a convex optimization problem and try to solve it. But before we go there, what I wanted to do is I wanted to illustrate some additional properties of convex functions which will, which we will actually use to prove certain functions in the machine learning world are actually convex. So, here are some additional properties which I wanted to discuss. And these properties will come in very handy when you are playing around with convex functions to prove certain functions are convex.

So, let us see some intuitive properties and there are a host of properties, laundry list of properties. I am not going to cover all of them in this course, but we will see a very few which are relevant and important in the context of machine learning. Additional, some additional properties, let me say some, because these are not necessarily all properties.

So, property 1. Some of these will be obvious, but I think it is important to go over a few of them so that we see how you can break down a potentially complicated function into pieces. And if

these, each piece is a convex function, then you can kind of use these properties to say when the, more complicated function is also convex. So, that is the idea.

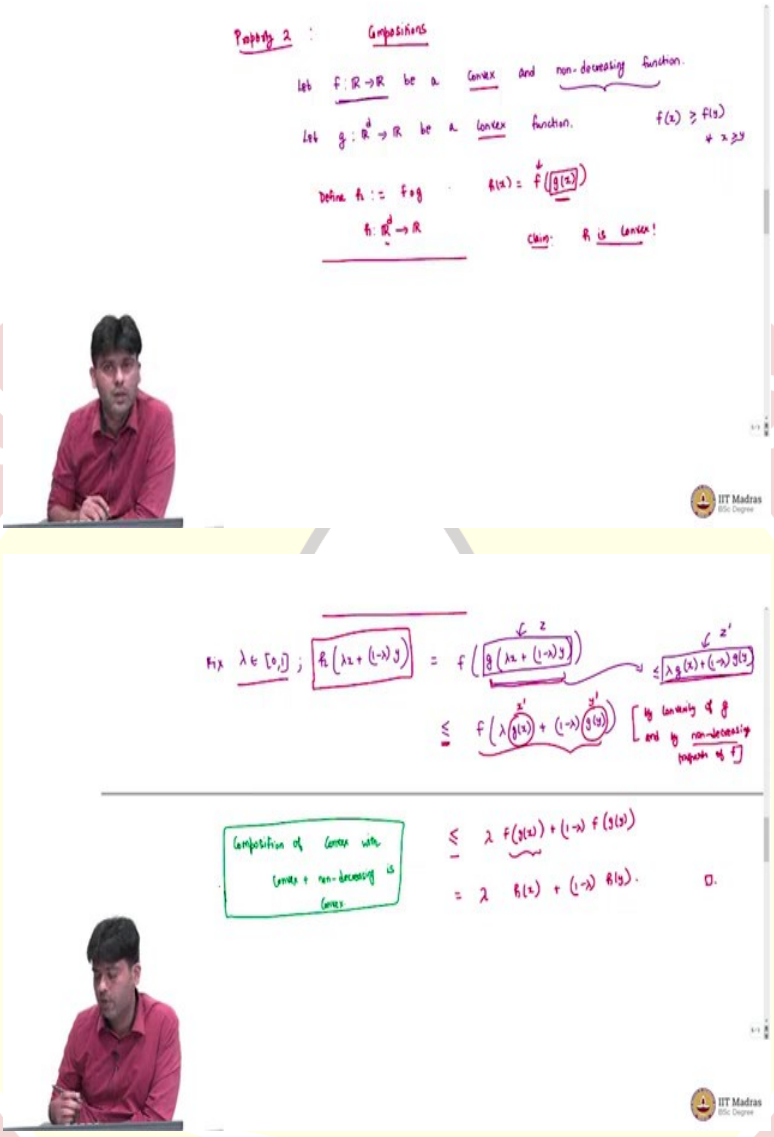
Property 1, if f, well, could be from Rd to R and g, I am just using Rd to R, but in general, this could be any domain, but for this course, I think, Rd to R would suffice, so are both convex functions then define h of x as f of x plus g of x, h of x as defined like this, is also a convex function. Why is this true? Pause for a second and think. This is a very easy proof. I will anyway do it. But you should already be able to see it if you have not already. So, I mean, if you think about it for a second, you will see why this is true.

Proof. Why is this true? Well, by definition, let us look at h of lambda x plus 1 minus lambda y for two points x and y. So, for any two points x and y, well, if this guy is convex, if h is convex, then what can we say, what should we be able to argue about this. Well, this is the functions value at lambda x plus 1 minus lambda y, now, this should be less than or equal to lambda times h of x plus 1 minus lambda times h of y. If we can prove that, then we are done. So, then it means that h of x is convex. But why is that true?

Well, by our definition of h this is simply f of lambda x plus 1 minus lambda y plus g of lambda x plus 1 minus lambda y. This is by definition of h. Now, f and g are convex. So, which means that the functions value of f is upper bounded by lambda f of x plus 1 minus lambda f of y plus lambda g of x plus 1 minus lambda g of y. This is by convexity of f and g. Now, this is simply lambda times f of x plus g of x plus 1 minus lambda times f of y plus g of y. So, this, of course, for all lambda fix lambda belonging to 0 comma 1 this is true.

Well, what is this? This is simply lambda h of x plus 1 minus lambda h of y. So, now, this proves that h of x is convex, because h of x now is less than or equal to lambda h of x plus 1 minus lambda h of y. There is nothing that we assumed about x and y, nothing we assumed about lambda except that it belongs to 0 or not. So, this is convex. So, what is the takeaway? The takeaway is that sums of convex functions is complex. This is property 1.

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We will see where we can use this in a bit. But for the moment let us go ahead and look at the other properties that I wanted to discuss. Property 2, this property is about compositions. So, you take two functions and compose them, which means, you apply these functions one after the other and then ask when is the composition of functions also convex. So, under what conditions can you guarantee that or what conditions are sufficient to say that the compositions are convex function. We will look at some examples.

So, first example is let f, we are going to think of f now as R to R be a what I will call as convex and non-decreasing function, which means it is an increasing function but it need not strictly increase, so it cannot decrease. By non-decreasing it simply means that if f is non-decreasing, then

f of x is greater than or equal to f of y for all x greater than or equal to y. So, f remember is R to R. There is no Rd. It is a simple real function. It is a convex function. It is a non-decreasing function. Let us, let that be f.

Now, here is another function g, which is Rd to R be a convex function. Now, what can we say about this new function h? So, define h as f of g of f composition g. So, by which what I mean h of x = f of g of x. So, what is the domain? So, it is Rd to R. This is a function from Rd to R, but then h operates like this. It takes Rd first and then applies g first to the Rd vector which would give a real value and then it applies f over there. That is the kind of function that we are looking at.

So, this g itself is Rd to R, f is then maps this R to R. We are assuming g is convex and f is convex and also non-decreasing. Now, the question is what can we say about the composition of f and g? So, the claim is h is convex and this is not hard to see also. Again, feel free to pause and try out why this should be true. I will do it anyway. So, why is this true? So, let us try to analyze why this is true.

Well, I am going to think of h of lambda x plus 1 minus lambda y. I want to consider this for some lambda in 0 comma 1 fix, some lambda 0 comma 1 and some x and y in Rd and then ask what is h of lambda x plus 1 minus lambda y. So, this by our definition of f of g of x is just f of g of lambda x plus 1 minus lambda y. Now, we know that g is a convex function.

So, which means that I can write g of lambda x plus 1 minus lambda y as at most by convexity of g I can write this as lambda g of x plus 1 minus lambda g of y, which means that if I input this value to f and I input this value f, the f of this box the value evaluated at g of lambda x plus 1 minus lambda y which is a real number is has to be less than f of lambda g of x plus 1 minus lambda g of y.

Why is that? That is simply because f is now a non-decreasing function. So, think of this value as z and this as z dash. We know that z is less than or equal to z dash which means that f of z has to be less than or equal to f of z dash. That is what non-decreasing of f means. So, which means that I can say that this whole thing is less than f of lambda g of x plus 1 minus lambda g of y. So, how did this happen? We have used two properties here. The first property we use is by convexity of g by which this happens and by non-decreasing property of f.

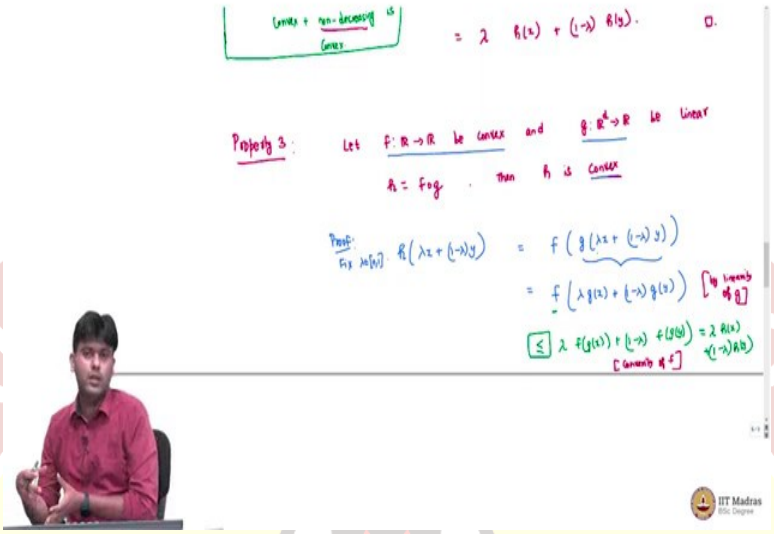
So, which is when I can say that this is less than or equal to this. Otherwise, if g, if f is not necessarily non-decreasing, I cannot say that this f of something is less than or equal to this. So, because though z which is g of lambda x plus 1 minus lambda y is less than or equal to lambda g of x plus 1 minus lambda g of y, f might give a higher value to f of z than f of z dash so if you do not assume anything about f. So, even if you assume it is convex that can still happen.

So, we kind of want to then say this f is non-decreasing and so this is true. Now, I will invoke convexity. And what would that mean? Well, now that means that this has to be less than, now look at this. So, g of x is a real number, g of y is a real number. Now, this is f of lambda times some real number plus 1 minus lambda times some other real number. So, think of this as some x dash and this as y dash. Now, by convexity of f we know that this has to be less than lambda times f of x dash which is g of x plus 1 minus lambda y dash which is g of y.

But then what is this. Well, this is simply h of x by our definition of h which is the composition of f and g plus 1 minus lambda h of y. So, which means what we have managed to show is that the h function evaluated at lambda x plus 1 minus lambda y is less than or equal to the lambda h of x plus 1 minus lambda h of y.

And again, because this can happen for any lambda between 0 and 1 and any x and y, this proves that f of g of x which is h of x is convex function. So, what is the takeaway? The takeaway is that composition of convex with convex plus non-decreasing is this is second property that we have come up with.

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Now, property 3. Now, one might want to ask, is this true? Is this non-decreasing nature of f really necessary? Can we say something when we simply do not assume anything about the non-decreasing nature of f. In general, you will not be able to say anything. But if g is special, then you do not really need non-decreasing nature of f. So, and that is property 3. So, our property 3 would be as follows.

Let f R to R be convex and g Rd to R be linear. So, I am assuming g not as a convex function, what as a linear function now. Now, again define h as f composition g. It is a composition of linear and a convex function, then h is convex. This is the claim. Now, I am, remember or note that I am only assuming f is convex, I am not imposing f to be non-decreasing. It is just convex, but then I have now changed g from a general convex function to a linear function. Then I want to argue that h is convex. Now, how will we do that?

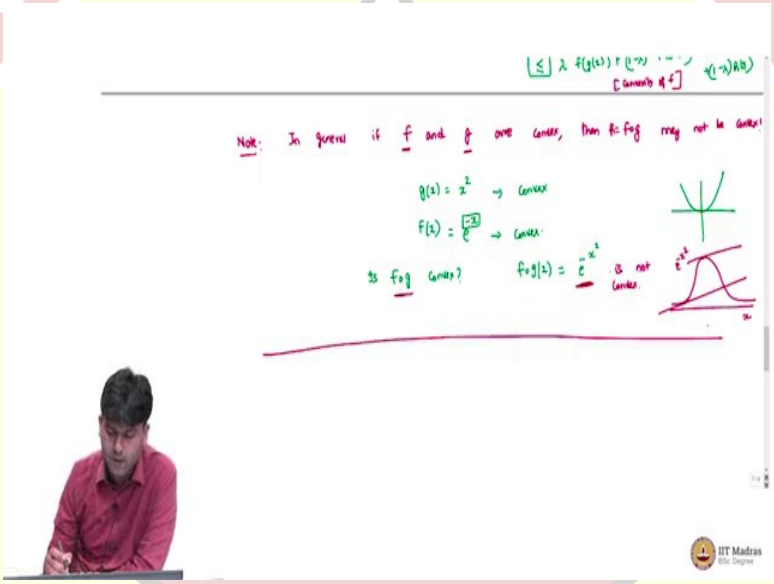
That is again straightforward. So, you can again use h of lambda x plus 1 minus lambda y again from the definition fix lambda 0 comma 1. Now, this is by definition f of g of lambda x plus 1 minus lambda y. Now, what can we say about g of lambda x plus 1 minus lambda y. Well, g is a linear function. If g is a linear function, then this, what can this value be, what will this value be?

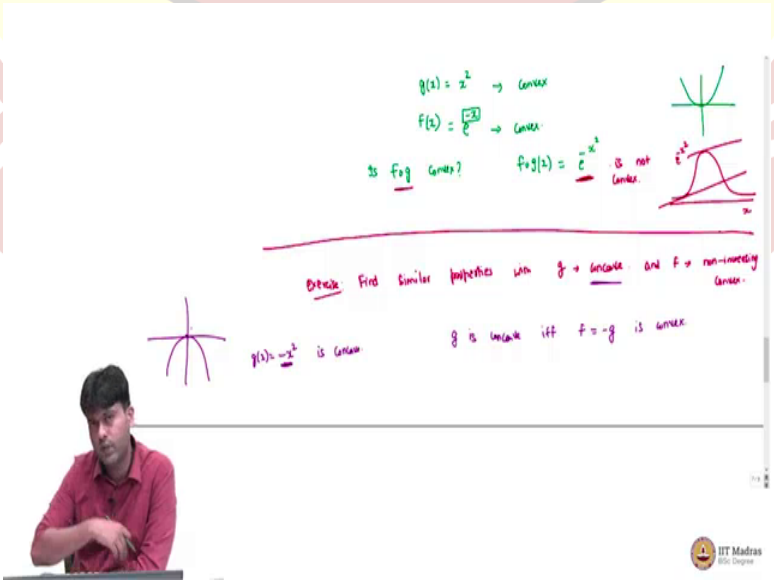
Well, if you linearly interpolate x and y, then the output will also get a linear interpolation. So, that is what the property of linear function is. So, this is simply equal to lambda g of x plus 1 minus lambda f, g of y this is an equality there. Why is this? This is by linearity of g. So, in general, we

are not assuming anything about f. This is by, this is exactly same as this. So, that is the linearity part.

Now, we invoke the convexity of f and then say that this is less than or equal to lambda f of g of x plus 1 minus lambda f of g of y, which is simply lambda h of x plus 1 minus lambda h of y. Now, this is again, let me make it very clear, convexity of f. So, you take a linear function and then compose it with a convex function what you get is a function that is convex. That is what this theorem says, this property, simple property says, which is good.

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So, here is an example where the general, in general, let me note that, in general, if f and g are convex then the composition h may not to be convex. So, this may not happen in general unless we assume something extra about f or g. So, the thing that we assumed about g in this case was it is linear in which case you do not have to assume anything about f it is still convex or if you do not assume anything about g just that it is convex then you have to assume f is non-decreasing then we can show this is convexity. In general, this may not be true. Here is an example to argue why this is the case.

Let us say our g of x is x squared. So, this is, we have already seen this is a simple convex function. So, this is simply complex. g of x is x square is convex. Now, the other thing that we want to argue is that we want to look at, let us look at f of x, well, let us think of f of x as e power minus x. Is this a convex function, is this convex?

Well, e power anything is a convex, I mean, e power minus x or e power minus x is a convex function, you can simply think of this as e power x is a convex function, but then this minus x is simply a linear function of x. So, we can apply our convex over linear to say that the e power minus x is convex, so convex.

So, now, let us ask if f of g of x is convex, is f of g of x convex? Now, what is f of g of x? Well, f of g of x is going to be, well, first I apply g, I will get x square, but then for whatever I evaluate, then I apply my e power minus, so this is going to be e power minus x square. Now, e power minus x square is not convex. I urge you to try to prove this from first principles why this is not convex using several definitions that we have already seen.

But the visually this, if you have encountered the famous Gaussian bell curve so that has a form e to the minus x square, where let us say the mean of the Gaussian PDF is 0 and let us say the variance is 1 and it is just a scaled version of this. So, it looks visually like this, something like this, e power minus x square.

Clearly this is at least visually this is not convex. So, why because, if we take any point and necessarily the tangent might not lower bound everybody, in fact, point here would actually have all the values lesser, so this is not convex. So, this is a very visual proof. But then I urge you to try it out from other principles that, other definitions that we have. But the point I wanted to illustrate

is that if f and g are both convex, it is not necessary that the f composition g is a convex function. So, this is property.

Now, I also urge you to try and play around with what else, what other ways of composition can you do. We just took convex linear, convex non-increasing, non-decreasing and so on, but you can argue something very similar with, find similar, so this is an exercise, find similar properties with f a concave function or g a concave function and f is non-increasing. So, think of what combination should f and g satisfy so that the composition becomes a convex function.

So, what is a concave function, just to be sure, we will call g a concave function, g is concave if and only if, this is a definition, f is, f equal to minus g is convex, so if the negative of that function is convex. So, for instance, minus x square is g of x equal to minus x square is concave. So, how does the picture look like? The picture would look like this. It is always going to be negative and it will have the highest value 0 at 0. This is a concave function. It is like an inverted convex function.